Step 2 – Propose High-Level Design and Get Buy-In

The high-level design of a news feed system is divided into two major flows:

1. Feed Publishing
2. News Feed Building

These two operations form the core of the entire system.

Two Main Flows

Flow	Purpose
Feed Publishing	Publish user posts and distribute them
News Feed Building	Retrieve and display feed posts

1. Feed Publishing

Feed publishing happens when a user creates a new post.

Main Goal

When a user publishes content:

- The post is stored
- Friends/followers receive the post in their feed
- Notifications may be triggered

Feed Publishing Workflow

The publishing process includes:

1. User creates a post
2. Data is written to cache and database
3. Fanout service distributes the post
4. Friends’ feeds are updated
5. Notifications are sent

2. News Feed Building

News feed building happens when a user opens the feed page.

Main Goal

Build a personalized feed containing:

- Friends’ posts
- Latest updates
- Reverse chronological ordering

Simplified Assumption

For simplicity:

```None
The feed is sorted in reverse chronological order.
```

Meaning:

- Newest posts appear first
- No advanced ranking algorithm is applied

News Feed APIs

Clients communicate with the backend using:

HTTP APIs

The APIs support actions such as:

- Publishing posts
- Retrieving feeds
- Adding friends
- Sending notifications

Two major APIs are discussed below.

1. Feed Publishing API

This API allows users to create posts.

API Endpoint

```
None

POST /v1/me/feed
```

Parameters

Parameter	Description
content	Text content of the post
auth_token	Used for authentication

Example Request

```
None

POST /v1/me/feed


content=Hello

auth_token={auth_token}
```

Purpose

This API:

- Authenticates the user
- Stores the post
- Triggers fanout process

2. News Feed Retrieval API

This API retrieves a user's feed.

API Endpoint

```None
GET /v1/me/feed
```

Parameters

Parameter	Description
auth_token	Used for authentication

Purpose

This API:

- Fetches feed posts
- Retrieves cached feed data
- Returns ordered news feed items

Feed Publishing High-Level Design

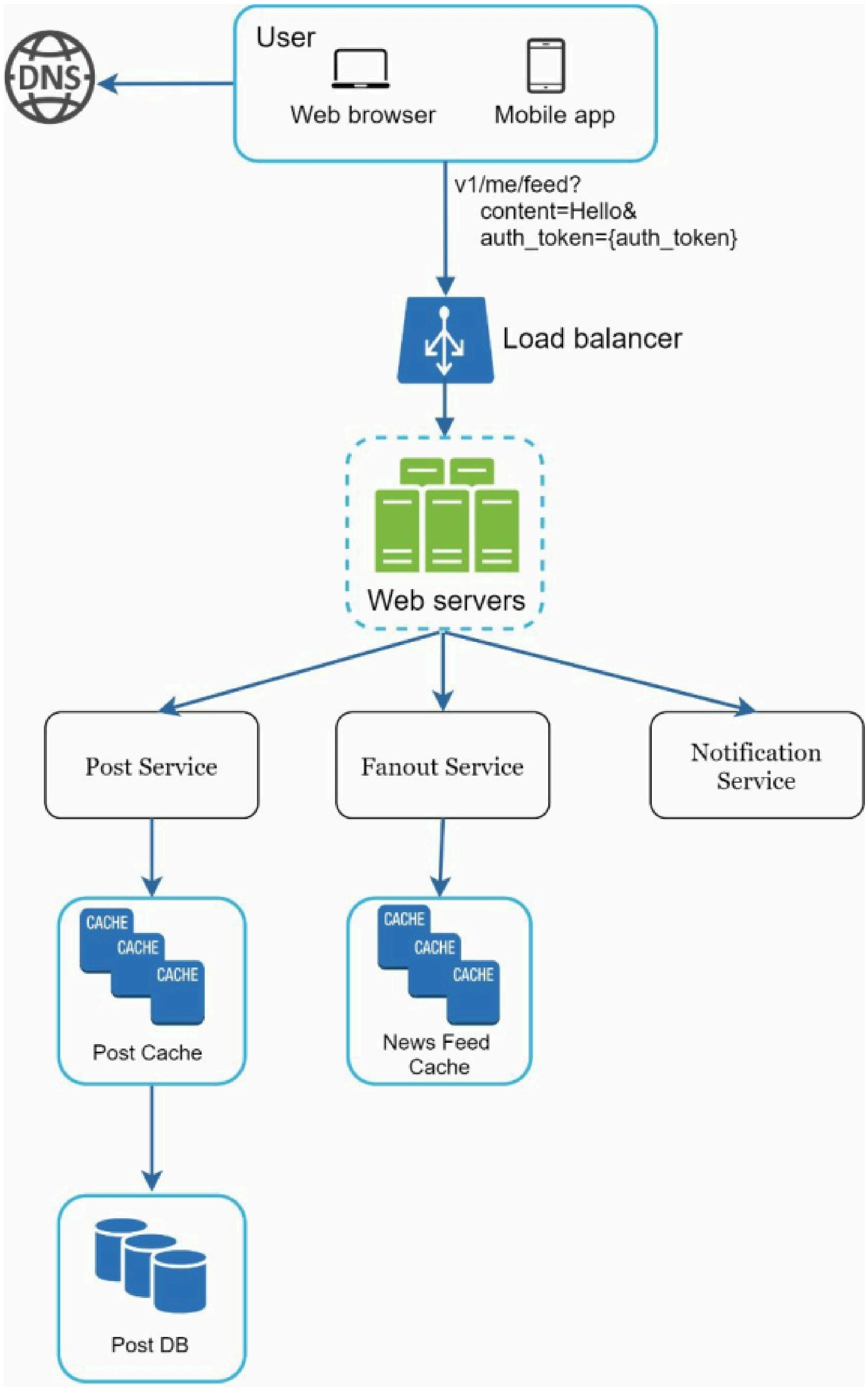

Figure shows the high-level architecture of feed publishing.

Components in Feed Publishing Flow

1. User

A user creates a new post using:

- Mobile app
- Web browser

Example Request

```None
/v1/me/feed?content=Hello&auth_token={auth_token}
```

2. Load Balancer

The load balancer:

- Receives incoming requests
- Distributes traffic across web servers

Purpose

- Improve scalability

- Prevent server overload
- Increase availability

3. Web Servers

Web servers:

- Handle incoming API requests
- Route traffic to internal services

Responsibilities

- Authentication
- Request validation
- Service routing

4. Post Service

The post service is responsible for:

Persisting Posts

Responsibilities

- Store posts in database
- Store recent posts in cache
- Generate post metadata

Storage Used

Storage	Purpose
Database	Permanent storage
Cache	Fast access

5. Fanout Service

The fanout service distributes posts to followers/friends.

Main Responsibility

```None
Push new content into friends' news feeds.
```

Purpose

Avoid building feeds repeatedly from scratch.

Feed Cache

Fanout service updates:

News Feed Cache

for fast feed retrieval later.

6. Notification Service

The notification service informs users about new posts.

Notifications Can Include

- Push notifications
- Mobile alerts
- In-app notifications

Example

```None
Your friend uploaded a new post.
```

Feed Publishing Summary

Component	Responsibility

User	Create post
Load Balancer	Distribute requests
Web Server	Route requests
Post Service	Store post
Fanout Service	Update followers’ feeds
Notification Service	Send alerts

News Feed Building High-Level Design

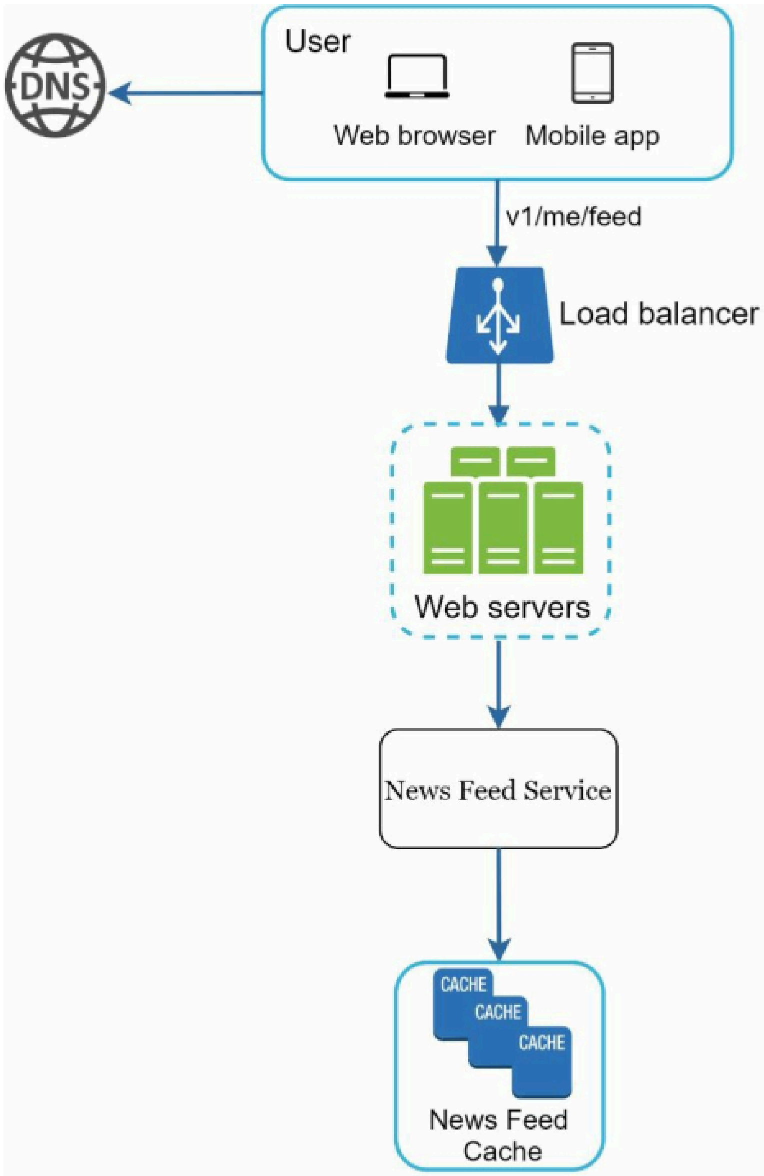

Figure shows how feeds are generated and retrieved.

Components in News Feed Retrieval Flow

1. User

The user opens the feed page.

Example Request

```None
GET /v1/me/feed
```

2. Load Balancer

The load balancer forwards requests to available web servers.

3. Web Servers

Web servers route requests to:

News Feed Service

4. News Feed Service

The news feed service:

- Fetches feed data
- Retrieves feed IDs
- Builds the response

Main Goal

Return feed content with:

- Low latency
- Correct ordering
- Personalized data

5. News Feed Cache

The cache stores:

```
None
Feed IDs and recent feed data required to render the news feed.
```

Why Cache is Important?

Without cache:

- Feed generation becomes slow
- Database load increases significantly

Benefits of Feed Cache

- Faster feed retrieval
- Reduced database load
- Better scalability
- Improved user experience

Overall High-Level Architecture

The complete high-level system supports:

Feature	Supported
Post publishing	Yes
Feed retrieval	Yes
Feed caching	Yes
Notifications	Yes
Scalability	Yes
Reverse chronological feed	Yes

Key Design Idea

The core design principle is:

```None
Separate feed publishing and feed retrieval flows for scalability and performance optimization.
```